**LECTURE 11**

**GMM I**

# Definition

Suppose that an econometrician observes data $\{W_i : i = 1, \ldots, n\}$, where $W_i$ is a random $p$-vector. Let $g$ be an $l$-dimensional function depending on $W_i$ and the $k$-vector of parameters $b$:

$$g(W_i, b) = \begin{pmatrix} g_1(W_i, b) \\ \vdots \\ g_l(W_i, b) \end{pmatrix},$$

and $g_j : \mathbb{R}^p \times \mathbb{R}^k \to \mathbb{R}$ for $j = 1, \ldots, l$. The model is defined by the following *moment condition*:

$$\mathrm{E}[g(W_i, \beta)] = 0 \text{ for some } \beta \in \mathbb{R}^k. \tag{1}$$

Examples:

- **Linear regression.** Let $W_i = \left(Y_i, X_i^\top\right)^\top$ and $Y_i = X_i^\top \beta + U_i$, where $\beta \in \mathbb{R}^k$, and $\mathrm{E}[X_i U_i] = 0$. In this case, $g(W_i, b) = X_i\left(Y_i - X_i^\top b\right)$, $l = k$, and the moment condition is $\mathrm{E}\left[X_i\left(Y_i - X_i^\top \beta\right)\right] = 0$.

- **IV regression.** Let $W_i = \left(Y_i, X_i^\top, Z_i^\top\right)^\top$, $Y_i = X_i^\top \beta + U_i$, where $\beta \in \mathbb{R}^k$, and $\mathrm{E}[Z_i U_i] = 0$, where $Z_i$ is an $l$-vector. In this case, $g(W_i, b) = Z_i\left(Y_i - X_i^\top b\right)$ with the moment condition $\mathrm{E}\left[Z_i\left(Y_i - X_i^\top \beta\right)\right] = 0$.

- **Lucas's model.** (This example uses time subscripts $t$ rather than $i$, reflecting the time-series nature of the data.) Suppose that in period $t$, investors derive utility from consumption $C_t$. Let $R_{j,t}$ be the rate of return on risky asset $j$. Suppose that there are $m$ assets. Assume that the utility function is $\sum_{t=1}^\infty \delta^t C_t^{1-\alpha}/(1-\alpha)$. In equilibrium, the returns on risky assets are determined by the following Euler equations:
$$\mathrm{E}\left[\delta\left(\frac{C_{t+1}}{C_t}\right)^{-\alpha}(1 + R_{j,t+1})\right] = 1, \quad j = 1, \ldots, m.$$
Here $W_t = (C_t, R_{1,t}, \ldots, R_{m,t})$, $b = (a, d)$, $g_j(W_t, b) = d\left(\frac{C_{t+1}}{C_t}\right)^{-a}(1 + R_{j,t+1}) - 1$ for $j = 1, \ldots, m$, and the moment conditions are given by the above equations. Here, $g$ is nonlinear in the parameters.

The model is *identified* if $\mathrm{E}[g(W_i, \beta)] = 0$ and $\mathrm{E}[g(W_i, \widetilde{\beta})] = 0$ imply $\beta = \widetilde{\beta}$, that is, the solution of (1) is unique. The moment condition gives us $l$ restrictions for $k$ parameters. A necessary condition for the model to be identified is that $l \geq k$, that is, there must be at least $k$ restrictions. This necessary condition is called the *order condition*. The model is *underidentified* if the order condition fails.

When $k = l$, applying the MM principle, we can estimate $\beta$ by the value of $b$ that solves the sample analogue of (1):

$$n^{-1}\sum_{i=1}^n g\left(W_i, \widehat{\beta}_n^{MM}\right) = 0.$$

When $l > k$, no $b \in \mathbb{R}^k$ generally solves all $l$ equations exactly. In this case, we can choose the value of $b$ that makes the sample moments as close to zero as possible. Let $A_n$ be a (possibly random) $l \times l$ weight matrix such that $A_n \to_p A$, where $A$ is non-random and has full rank $l$. The *Generalized Method of Moments (GMM) estimator* of $\beta$ is defined to be the value of $b$ that minimizes the weighted distance of $n^{-1}\sum_{i=1}^n g(W_i, b)$

from zero:

$$\widehat{\beta}_n^{GMM} = \arg\min_{b\in B} \left\| A_n n^{-1} \sum_{i=1}^n g\left(W_i, b\right) \right\|^2$$
$$= \arg\min_{b\in B} \left( n^{-1} \sum_{i=1}^n g\left(W_i, b\right) \right)^\top A_n^\top A_n \left( n^{-1} \sum_{i=1}^n g\left(W_i, b\right) \right). \tag{2}$$

The set $B \subset \mathbb{R}^k$ is assumed to be compact. The matrix $A^\top A$ is positive definite.

## Linear case

In this section, we discuss the IV regression example in detail. Here $g$ is linear in the parameters. As in Lecture 10, we assume that some or all of the $k$ regressors in $X_i$ are endogenous:

$$\mathrm{E}[X_i U_i] \neq 0,$$

and that the $l$ instruments $Z_i$ are weakly exogenous:

$$\mathrm{E}[Z_i U_i] = 0.$$

The model is identified if the following *rank condition* is satisfied:

$$\operatorname{rank}\left(\mathrm{E}\left[Z_i X_i^\top\right]\right) = k.$$

If the rank condition is satisfied and $l = k$, we say that the model is *exactly* or *just* identified. We say that the model is *overidentified* if the rank condition is satisfied and $l > k$ (there are more instruments than parameters). Unlike in Lecture 10, the model may be overidentified here.

In the linear IV regression case, $\widehat{\beta}_n^{GMM}$ is the minimizer of

$$\left( n^{-1} \sum_{i=1}^n Z_i \left(Y_i - X_i^\top b\right) \right)^\top A_n^\top A_n \left( n^{-1} \sum_{i=1}^n Z_i \left(Y_i - X_i^\top b\right) \right),$$

and is given by the following expression:

$$\widehat{\beta}_n^{GMM} = \left( \sum_{i=1}^n X_i Z_i^\top \left(A_n^\top A_n\right) \sum_{i=1}^n Z_i X_i^\top \right)^{-1} \sum_{i=1}^n X_i Z_i^\top \left(A_n^\top A_n\right) \sum_{i=1}^n Z_i Y_i.$$

We show next that the GMM estimator is consistent. We need the following assumptions.

- $\{(Y_i, X_i, Z_i) : i = 1, \ldots, n\}$ are iid.
- $Y_i = X_i^\top \beta + U_i$, where $\beta \in \mathbb{R}^k$.
- $\mathrm{E}\left[Z_i U_i\right] = 0$.
- $\mathrm{E}\left[Z_i X_i^\top\right]$ has rank $k$.
- $A_n \to_p A$, where $A$ has rank $l \geq k$.
- $\mathrm{E}[X_{i,j}^2] < \infty$ for all $j = 1, \ldots, k$.
- $\mathrm{E}[Z_{i,j}^2] < \infty$ for all $j = 1, \ldots, l$.

Write

$$\widehat{\beta}_n^{GMM} = \beta + \left(n^{-1}\sum_{i=1}^n X_i Z_i^\top \left(A_n^\top A_n\right) n^{-1}\sum_{i=1}^n Z_i X_i^\top\right)^{-1} n^{-1}\sum_{i=1}^n X_i Z_i^\top \left(A_n^\top A_n\right) n^{-1}\sum_{i=1}^n Z_i U_i.$$

The last two assumptions imply that

$$\mathrm{E}\left|X_{i,r}Z_{i,s}\right| < \infty \quad \text{for all } r = 1,\dots,k \text{ and } s = 1,\dots,l.$$

By the WLLN,

$$n^{-1}\sum_{i=1}^n X_i Z_i^\top \to_p \mathrm{E}[X_i Z_i^\top].$$

Since $A_n \to_p A$, we also have that

$$n^{-1}\sum_{i=1}^n X_i Z_i^\top \left(A_n^\top A_n\right) n^{-1}\sum_{i=1}^n Z_i X_i^\top \to_p \mathrm{E}[X_i Z_i^\top]\left(A^\top A\right)\mathrm{E}[Z_i X_i^\top].$$

Further, since $\mathrm{E}\left[Z_i X_i^\top\right]$ has rank $k$ and $A$ has rank $l \geq k$, it follows that the $k\times k$ matrix $\mathrm{E}[X_i Z_i^\top]\left(A^\top A\right)\mathrm{E}[Z_i X_i^\top]$ has full rank $k$ and is therefore invertible. Consequently, by Slutsky's Theorem,

$$\left(n^{-1}\sum_{i=1}^n X_i Z_i^\top \left(A_n^\top A_n\right) n^{-1}\sum_{i=1}^n Z_i X_i^\top\right)^{-1} \to_p \left(\mathrm{E}[X_i Z_i^\top]\left(A^\top A\right)\mathrm{E}[Z_i X_i^\top]\right)^{-1}.$$

Next, by the WLLN,

$$n^{-1}\sum_{i=1}^n Z_i U_i \to_p 0,$$

and thus $\widehat{\beta}_n^{GMM} \to_p \beta$.

To show asymptotic normality, we will need the following three assumptions in addition to the above.

- $\mathrm{E}[Z_{i,j}^4] < \infty$ for all $j = 1,\dots,l$.
- $\mathrm{E}[U_i^4] < \infty$.
- $\mathrm{E}[U_i^2 Z_i Z_i^\top]$ is positive definite.

Write

$$n^{1/2}\left(\widehat{\beta}_n^{GMM} - \beta\right) = \left(n^{-1}\sum_{i=1}^n X_i Z_i^\top \left(A_n^\top A_n\right) n^{-1}\sum_{i=1}^n Z_i X_i^\top\right)^{-1} n^{-1}\sum_{i=1}^n X_i Z_i^\top \left(A_n^\top A_n\right) n^{-1/2}\sum_{i=1}^n Z_i U_i.$$

The last two assumptions imply that the variance of $Z_i U_i$, $\mathrm{E}[U_i^2 Z_i Z_i^\top]$, is finite. By the CLT,

$$n^{-1/2}\sum_{i=1}^n Z_i U_i \to_d N\left(0, \mathrm{E}[U_i^2 Z_i Z_i^\top]\right).$$

Define

$$\begin{aligned} Q &= \mathrm{E}[Z_i X_i^\top], \\ \Omega &= \mathrm{E}[U_i^2 Z_i Z_i^\top]. \end{aligned}$$

Combining the above results, we have

$$n^{1/2}\left(\widehat{\beta}_n^{GMM} - \beta\right) \to_d N\left(0, V\right),$$

where $V$ takes the sandwich form:

$$V = \left(Q^\top A^\top AQ\right)^{-1} Q^\top A^\top A\Omega A^\top AQ \left(Q^\top A^\top AQ\right)^{-1}.$$

The variance-covariance matrix $V$ can be estimated by replacing $A$, $Q$, and $\Omega$ with their consistent estimators $A_n$, $\widehat{Q}_n$, and $\widehat{\Omega}_n$, respectively, where

$$\widehat{Q}_n = n^{-1}\sum_{i=1}^{n} Z_i X_i^\top,$$

$$\widehat{\Omega}_n = n^{-1}\sum_{i=1}^{n} \widehat{U}_i^2 Z_i Z_i^\top,$$

and $\widehat{U}_i = Y_i - X_i^\top \widehat{\beta}_n^{GMM}$.

## General case

In the general case, the GMM estimator minimizes the nonlinear function in (2). Usually, we do not have a closed-form expression for $\widehat{\beta}_n^{GMM}$, and the minimization must be done using numerical procedures. Nevertheless, under general regularity conditions, it is possible to show that $\widehat{\beta}_n^{GMM}$ is consistent and asymptotically normal. We provide heuristic arguments for consistency and asymptotic normality.

Since the criterion function in (2) involves averages, we should expect that

$$\left\| A_n n^{-1}\sum_{i=1}^{n} g\left(W_i, b\right)\right\|^2 \to_p \left\|A\,\mathrm{E}[g\left(W_i, b\right)]\right\|^2. \tag{3}$$

Assuming that the model is uniquely identified, $\mathrm{E}[g\left(W_i, b\right)] = 0$ if and only if $b = \beta$. Since $\left\|A\,\mathrm{E}[g\left(W_i, b\right)]\right\|^2 > 0$ for all $b \neq \beta$, the true value $\beta$ is the unique minimizer of $\left\|A\,\mathrm{E}[g\left(W_i, b\right)]\right\|^2$. Intuitively, $\widehat{\beta}_n^{GMM}$ is consistent because

$$\begin{aligned}
\widehat{\beta}_n^{GMM} &= \arg\min_{b\in B} \left\| A_n n^{-1}\sum_{i=1}^{n} g\left(W_i, b\right)\right\|^2 \\
&\to_p \arg\min_{b\in B} \left\|A\,\mathrm{E}[g\left(W_i, b\right)]\right\|^2 \\
&= \beta.
\end{aligned}$$

The formal proof of consistency requires regularity conditions: (i) compactness of $B$, (ii) continuity of $g(w, b)$ in $b$ for each $w$, (iii) *uniform* convergence of the criterion over $b \in B$ in (3), and (iv) unique identification, that is, $\beta$ is the unique minimizer of the population criterion.

For asymptotic normality, $\widehat{\beta}_n^{GMM}$ solves the first-order conditions:

$$\left(n^{-1}\sum_{i=1}^{n} \frac{\partial g\left(W_i, \widehat{\beta}_n^{GMM}\right)}{\partial b^\top}\right)^\top A_n^\top A_n n^{-1}\sum_{i=1}^{n} g\left(W_i, \widehat{\beta}_n^{GMM}\right) = 0. \tag{4}$$

In fact, it is sufficient if $\widehat{\beta}_n^{GMM}$ solves the first-order conditions approximately: the right-hand side of (4) may be $o_p(n^{-1/2})$ rather than exactly zero. This accommodates numerical optimization where the solution is found up to a tolerance that shrinks faster than $n^{-1/2}$.

Next, using the expansion of $g\left(W_i, \widehat{\beta}_n^{GMM}\right)$ around $g\left(W_i, \beta\right)$ (the element-by-element mean value theorem), we obtain

$$g\left(W_i, \widehat{\beta}_n^{GMM}\right) = g\left(W_i, \beta\right) + \frac{\partial g\left(W_i, \widehat{\beta}_n^{*}\right)}{\partial b^\top}\left(\widehat{\beta}_n^{GMM} - \beta\right), \tag{5}$$

where $\widehat{\beta}_n^*$ is between $\widehat{\beta}_n^{GMM}$ and $\beta$. Substitution of (5) into (4) gives

$$0 = \left(n^{-1}\sum_{i=1}^{n}\frac{\partial g\left(W_i,\widehat{\beta}_n^{GMM}\right)}{\partial b^\top}\right)^\top A_n^\top A_n n^{-1}\sum_{i=1}^{n} g\left(W_i,\beta\right)$$

$$+\left(n^{-1}\sum_{i=1}^{n}\frac{\partial g\left(W_i,\widehat{\beta}_n^{GMM}\right)}{\partial b^\top}\right)^\top A_n^\top A_n\left(n^{-1}\sum_{i=1}^{n}\frac{\partial g\left(W_i,\widehat{\beta}_n^{*}\right)}{\partial b^\top}\right)$$

We can write

$$n^{1/2}\left(\widehat{\beta}\right.$$

$$= -\left(\left(n^{-1}\sum_{i=1}^{n}\frac{\partial g\left(W_i,\widehat{\beta}_n^{GMM}\right)}{\partial b^\top}\right)^\top A_n^\top A_n\left(n^{-1}\sum_{i=1}^{n}\frac{\partial g\left(W_i,\widehat{\beta}_n^{*}\right)}{\partial b^\top}\right)\right)^{-1}$$

$$\times\left(n^{-1}\sum_{i=1}^{n}\frac{\partial g\left(W_i,\widehat{\beta}_n^{GMM}\right)}{\partial b^\top}\right)^\top A_n^\top A_n n^{-1/2}\sum_{i=1}^{n}$$

Since $\mathrm{E}[g(W_i,\beta)]=0$, under regularity conditions,

$$n^{-1/2}\sum_{i=1}^{n} g\left(W_i,\beta\right)\to_d N\left(0,\mathrm{E}[g\left(W_i,\beta\right)g\left(W_i,\beta\right)^\top]\right).$$

(The asymptotic variance depends on the unknown $\beta$.) Since $\widehat{\beta}_n^{GMM}$ is consistent, and, as a result, $\widehat{\beta}_n^*\to_p\beta$ as well, under regularity conditions,

$$n^{-1}\sum_{i=1}^{n}\frac{\partial g\left(W_i,\widehat{\beta}_n^{GMM}\right)}{\partial b^\top}\to_p \mathrm{E}\left[\frac{\partial g\left(W_i,\beta\right)}{\partial b^\top}\right],$$

$$n^{-1}\sum_{i=1}^{n}\frac{\partial g\left(W_i,\widehat{\beta}_n^{*}\right)}{\partial b^\top}\to_p \mathrm{E}\left[\frac{\partial g\left(W_i,\beta\right)}{\partial b^\top}\right],$$

and that the matrix

$$\left(\mathrm{E}\left[\frac{\partial g\left(W_i,\beta\right)}{\partial b^\top}\right]\right)^\top A^\top A\left(\mathrm{E}\left[\frac{\partial g\left(W_i,\beta\right)}{\partial b^\top}\right]\right)$$

is invertible. Then,

$$n^{1/2}\left(\widehat{\beta}_n^{GMM}-\beta\right)\to_d N\left(0,V\right),$$

where

$$\begin{aligned}
V &= \left(Q^\top A^\top AQ\right)^{-1}Q^\top A^\top A\Omega A^\top AQ\left(Q^\top A^\top AQ\right)^{-1},\\
Q &= \mathrm{E}\left[\frac{\partial g\left(W_i,\beta\right)}{\partial b^\top}\right],\\
\Omega &= \mathrm{E}[g\left(W_i,\beta\right)g\left(W_i,\beta\right)^\top].
\end{aligned}$$

The variance-covariance matrix $V$ can be estimated by replacing $A$, $Q$, and $\Omega$ with their consistent estimators $A_n$, $\widehat{Q}_n$, and $\widehat{\Omega}_n$, respectively, where

$$\widehat{Q}_n = n^{-1} \sum_{i=1}^{n} \frac{\partial g\left(W_i, \widehat{\beta}_n^{GMM}\right)}{\partial b^\top},$$

$$\widehat{\Omega}_n = n^{-1} \sum_{i=1}^{n} g\left(W_i, \widehat{\beta}_n^{GMM}\right) g\left(W_i, \widehat{\beta}_n^{GMM}\right)^\top.$$